

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 25

Design of IIR Filters — Bilinear Transformation & Matched Z-Transform

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of the Bilinear Transformation (BLT)

1.1 Derivation from the Trapezoidal Integration Rule

The **Bilinear Transformation (BLT)** is an algebraic mapping derived by approximating continuous integration via numerical trapezoidal integration. Consider an ideal analog integrator:

$$H_a(s) = \frac{Y(s)}{X(s)} = \frac{1}{s} \iff \frac{dy(t)}{dt} = x(t) \quad (1)$$

Integrating both sides between the sampling instants $t = (n-1)T$ and $t = nT$:

$$y(nT) - y((n-1)T) = \int_{(n-1)T}^{nT} x(t) dt \quad (2)$$

Applying the **trapezoidal rule** to approximate the area under the curve:

$$\int_{(n-1)T}^{nT} x(t) dt \approx \frac{T}{2} [x(nT) + x((n-1)T)] \quad (3)$$

Substituting this into the discrete difference equation ($y[n] \equiv y(nT)$):

$$y[n] - y[n-1] = \frac{T}{2} [x[n] + x[n-1]] \quad (4)$$

Taking the Z -transform on both sides:

$$\begin{aligned} Y(z) - z^{-1}Y(z) &= \frac{T}{2} [X(z) + z^{-1}X(z)] \\ Y(z)(1 - z^{-1}) &= \frac{T}{2} X(z)(1 + z^{-1}) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{T}{2} \left(\frac{1 + z^{-1}}{1 - z^{-1}} \right) \end{aligned} \quad (5)$$

Equating the analog integrator $H_a(s) = \frac{1}{s}$ to the discrete integrator $H(z)$:

$$\frac{1}{s} = \frac{T}{2} \left(\frac{1 + z^{-1}}{1 - z^{-1}} \right) \implies s = \frac{2}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) = \frac{2}{T} \left(\frac{z - 1}{z + 1} \right) \quad (6)$$

This is the celebrated **Bilinear Transformation substitution rule**.

1.2 Stability & The Elimination of Aliasing

- **Guaranteed Stability:** The entire open left-half of the s -plane ($\text{Re}(s) < 0$) maps strictly to the interior of the unit circle ($|z| < 1$). Stable analog filters **always produce stable digital filters**.
- **Zero Aliasing:** The entire continuous imaginary axis $s = j\Omega$ from $-\infty$ to $+\infty$ maps ****one-to-one**** onto the discrete unit circle $z = e^{j\omega}$ from $-\pi$ to $+\pi$.

- **Universal Applicability:** Because there is no aliasing, BLT can design **Lowpass, Highpass, Bandpass, and Bandstop filters**!

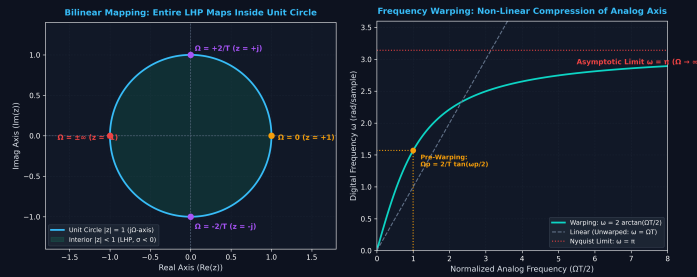


Figure 1: Left: One-to-one mapping from s -plane to z -plane. Right: Non-linear frequency warping curve showing compression into the digital interval $[0, \pi]$.

2 Frequency Warping & Mandatory Pre-Warping

2.1 Step-by-Step Derivation of the Frequency Warping Equation

To find the relationship between analog frequency Ω and digital frequency ω , evaluate the BLT substitution (6) on the unit circle $z = e^{j\omega}$ ($s = j\Omega$):

$$\begin{aligned} j\Omega &= \frac{2}{T} \left(\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right) = \frac{2}{T} \left[\frac{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2}(e^{j\omega/2} + e^{-j\omega/2})} \right] \\ &= \frac{2}{T} \left(\frac{e^{j\omega/2} - e^{-j\omega/2}}{e^{j\omega/2} + e^{-j\omega/2}} \right) \end{aligned} \quad (7)$$

Using Euler's identities $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ and $e^{j\theta} + e^{-j\theta} = 2 \cos \theta$:

$$j\Omega = \frac{2}{T} \left[\frac{2j \sin(\omega/2)}{2 \cos(\omega/2)} \right] = j \frac{2}{T} \tan \left(\frac{\omega}{2} \right) \quad (8)$$

Equating imaginary parts yields the fundamental **Frequency Warping Equation**:

$$\Omega = \frac{2}{T} \tan \left(\frac{\omega}{2} \right) \iff \omega = 2 \arctan \left(\frac{\Omega T}{2} \right) \quad (9)$$

The entire infinite analog axis $\Omega \in [0, \infty)$ is non-linearly compressed into the finite digital interval $\omega \in [0, \pi)$.

2.2 The Mandatory Pre-Warping Rule

Because frequency warping non-linearly shifts transition edges, any design specification given in digital frequencies ω_p, ω_s **must be pre-warped** into analog frequencies before designing the analog prototype filter:

$$\Omega_p = \frac{2}{T} \tan \left(\frac{\omega_p}{2} \right), \quad \Omega_s = \frac{2}{T} \tan \left(\frac{\omega_s}{2} \right) \quad (10)$$

3 The Matched Z-Transform (MZT) Method

The **Matched Z-Transform** maps poles and zeros individually via direct exponential substitution:

$$(s - s_k) \longrightarrow (1 - e^{s_k T} z^{-1}) \quad (11)$$

- **Poles:** $s = p_k \implies z_k = e^{p_k T}$.
- **Zeros at Infinity:** For an all-pole analog filter with N poles and no finite zeros, add $(1 + z^{-1})$ factors in the numerator to ensure transmission nulls at Nyquist ($\omega = \pi, z = -1$):

$$H(z) = K_d \frac{(1 + z^{-1})^M}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})} \quad (12)$$

- **Gain Constant K_d :** Evaluated to match DC gain: $H(1) = H_a(0)$.

4 Master Comparison Table of IIR Design Methods

Table 1: Comparison of Impulse Invariance, Bilinear Transformation, and Matched Z-Transform

Feature	Impulse Invariance	Bilinear Transformation (BLT)	Matched Z-Transform (MZT)
Substitution Rule	Partial fractions: $\frac{A}{s-s_k} \rightarrow \frac{TA}{1-e^{s_k T} z^{-1}}$	$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$ into $H_a(s)$	$(s-a) \rightarrow (1-e^{aT} z^{-1})$
Aliasing	**Severe aliasing**	**Zero aliasing**	Mild aliasing
Pre-Warping	Not applicable	**Mandatory** ($\Omega = \frac{2}{T} \tan(\omega/2)$)	Not applicable
Filter Types	LPF and BPF only (No HPF/BSF)	**Universal (LPF, HPF, BPF, BSF)**	LPF and BPF primarily
Stability	Strictly preserved	Strictly preserved	Strictly preserved

5 Standard Solved Exam Examples

Solved Exam Example 1: 2nd-Order Butterworth Lowpass Filter via BLT

Problem 1: Design a 2nd-order digital Butterworth lowpass filter with a 3-dB cutoff frequency $\omega_c = 0.2\pi$ rad/sample using the Bilinear Transformation with sampling interval $T = 1$ second.

Solution:

Step 1: Pre-Warp the Digital Cutoff Frequency Using the fundamental pre-warping relation:

$$\begin{aligned} \Omega_c &= \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = \frac{2}{1} \tan\left(\frac{0.2\pi}{2}\right) = 2 \tan(0.1\pi) = 2 \tan(18^\circ) \\ &= 2 \times 0.32492 = \mathbf{0.6498} \text{ rad/s} \end{aligned} \quad (13)$$

Step 2: Construct the Analog Prototype Transfer Function A normalized 2nd-order Butterworth lowpass filter has denominator $s^2 + \sqrt{2}s + 1$. Scaling by the pre-warped cutoff $\Omega_c = 0.6498$:

$$H_a(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\Omega_c s + \Omega_c^2} \quad (14)$$

Evaluating coefficients with all intermediate arithmetic:

$$\begin{aligned} \Omega_c^2 &= (0.64984)^2 = \mathbf{0.4223} \\ \sqrt{2}\Omega_c &= 1.41421 \times 0.64984 = \mathbf{0.9190} \\ H_a(s) &= \frac{0.4223}{s^2 + 0.9190s + 0.4223} \end{aligned} \quad (15)$$

Step 3: Substitute the Bilinear Transformation ($T = 1$ s) Substitute $s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$:

$$H(z) = \frac{0.4223}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 0.9190 \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right] + 0.4223} \quad (16)$$

Multiply the numerator and denominator by the common denominator $(1 + z^{-1})^2$:

$$\begin{aligned}\text{Numerator} &= 0.4223(1 + z^{-1})^2 = \mathbf{0.4223(1 + 2z^{-1} + z^{-2})} \\ \text{Denominator} &= 4(1 - z^{-1})^2 + 1.8380(1 - z^{-1})(1 + z^{-1}) + 0.4223(1 + z^{-1})^2 \\ &= 4(1 - 2z^{-1} + z^{-2}) + 1.8380(1 - z^{-2}) + 0.4223(1 + 2z^{-1} + z^{-2})\end{aligned}\quad (17)$$

Collecting like powers of z^{-1} step-by-step:

$$\begin{aligned}\text{Constant } (z^0) &= 4(1) + 1.8380(1) + 0.4223(1) = 4 + 1.8380 + 0.4223 = \mathbf{6.2603} \\ \text{Coefficient of } z^{-1} &= 4(-2) + 1.8380(0) + 0.4223(2) = -8 + 0.8446 = \mathbf{-7.1554} \\ \text{Coefficient of } z^{-2} &= 4(1) + 1.8380(-1) + 0.4223(1) = 4 - 1.8380 + 0.4223 = \mathbf{2.5843}\end{aligned}\quad (18)$$

The denominator is $D(z) = 6.2603 - 7.1554z^{-1} + 2.5843z^{-2}$. Divide all numerator and denominator terms by 6.2603 to make the denominator monic:

$$\begin{aligned}b_0 &= \frac{0.4223}{6.2603} = \mathbf{0.06746 \approx 0.0675} \\ a_1 &= -\frac{7.1554}{6.2603} = \mathbf{-1.14298 \approx -1.1430}, \quad a_2 = \frac{2.5843}{6.2603} = \mathbf{0.41281 \approx 0.4128}\end{aligned}\quad (19)$$

The final digital filter system transfer function is:

$$\mathbf{H(z) = \frac{0.0675(1 + 2z^{-1} + z^{-2})}{1 - 1.1430z^{-1} + 0.4128z^{-2}}}\quad (20)$$

Step 4: Comprehensive Frequency Response Verification

- **At DC** ($\omega = 0, z = 1$):

$$H(1) = \frac{0.0675(1 + 2(1) + 1)}{1 - 1.1430(1) + 0.4128(1)} = \frac{0.0675(4)}{0.2698} = \frac{0.2700}{0.2698} = \mathbf{1.0007 \approx 1.0} \quad (\text{ideal passband})\quad (21)$$

- **At Nyquist** ($\omega = \pi, z = -1$):

$$H(-1) = \frac{0.0675(1 - 2 + 1)}{1 + 1.1430 + 0.4128} = \frac{0}{2.5558} = \mathbf{0.0000} \quad (\text{exact transmission zero})\quad (22)$$

- **Stability Check:** Denominator roots have radius $r = \sqrt{a_2} = \sqrt{0.4128} \approx 0.6425 < 1$. Both poles lie inside the unit circle; filter is unconditionally stable.

Solved Exam Example 2: Highpass Filter Design via BLT

Problem 2: Design a first-order digital highpass filter with a 3-dB cutoff frequency $\omega_c = 0.5\pi$ rad/sample using the Bilinear Transformation with $T = 1$ second.

Solution:

Step 1: Pre-Warping

$$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 2 \tan\left(\frac{0.5\pi}{2}\right) = 2 \tan\left(\frac{\pi}{4}\right) = 2(1.0) = \mathbf{2.0 \text{ rad/s}}\quad (23)$$

Step 2: Analog Prototype A first-order highpass analog filter prototype is $H_a(s) = \frac{s}{s + \Omega_c} = \frac{s}{s + 2}$.

Step 3: BLT Substitution ($s = 2\frac{1-z^{-1}}{1+z^{-1}}$)

$$\begin{aligned}H(z) &= \frac{2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}{2\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2} = \frac{2(1 - z^{-1})}{2(1 - z^{-1}) + 2(1 + z^{-1})} \\ &= \frac{2(1 - z^{-1})}{2 - 2z^{-1} + 2 + 2z^{-1}} = \frac{2(1 - z^{-1})}{4} = \mathbf{0.5(1 - z^{-1})}\end{aligned}\quad (24)$$

Step 4: Frequency Response Checks

- **At DC** ($z = 1$): $H(1) = 0.5(1 - 1) = \mathbf{0.0}$ (complete blocking of low frequencies).
- **At Nyquist** ($z = -1$): $H(-1) = 0.5(1 - (-1)) = 0.5(2) = \mathbf{1.0}$ (lossless high-frequency transmission).
- **At 3-dB Cutoff** ($\omega = \pi/2, z = e^{j\pi/2} = j$):

$$H(j) = 0.5(1 - j^{-1}) = 0.5(1 + j) \implies |H(j)| = 0.5\sqrt{1^2 + 1^2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \approx \mathbf{0.7071} \quad \checkmark \quad (25)$$

Solved Exam Example 3: Matched Z-Transform (MZT) Application

Problem 3: Convert the analog bandpass section $H_a(s) = \frac{s+1}{(s+2)(s+3)}$ into an equivalent digital filter using the Matched Z-Transform with sampling period $T = 0.1$ second.

Solution:

1. **Map Finite Zeros and Poles** ($z_k = e^{s_k T}$):

$$\begin{aligned} \text{Zero at } s_0 = -1 &\implies z_0 = e^{-1(0.1)} = e^{-0.1} \approx \mathbf{0.9048} \\ \text{Pole at } s_1 = -2 &\implies p_1 = e^{-2(0.1)} = e^{-0.2} \approx \mathbf{0.8187} \\ \text{Pole at } s_2 = -3 &\implies p_2 = e^{-3(0.1)} = e^{-0.3} \approx \mathbf{0.7408} \end{aligned} \quad (26)$$

2. **Form System Function with Gain Constant** K_d :

$$H(z) = K_d \frac{1 - z_0 z^{-1}}{(1 - p_1 z^{-1})(1 - p_2 z^{-1})} = K_d \frac{1 - 0.9048 z^{-1}}{(1 - 0.8187 z^{-1})(1 - 0.7408 z^{-1})} \quad (27)$$

3. **Match DC Gain** ($z = 1 \iff s = 0$):

$$\begin{aligned} H_a(0) &= \frac{0 + 1}{(0 + 2)(0 + 3)} = \frac{1}{6} \approx \mathbf{0.1667} \\ \text{At } z = 1 : &\frac{1 - 0.9048}{(1 - 0.8187)(1 - 0.7408)} = \frac{0.0952}{(0.1813)(0.2592)} = \frac{0.0952}{0.0470} \approx \mathbf{2.0255} \\ K_d &= \frac{0.1667}{2.0255} \approx \mathbf{0.0823} \end{aligned} \quad (28)$$

4. **Final MZT Transfer Function:**

$$\mathbf{H(z)} = \frac{\mathbf{0.0823(1 - 0.9048z^{-1})}}{\mathbf{1 - 1.5595z^{-1} + 0.6065z^{-2}}} \quad (29)$$

6 Essential Exam Checklist & Traps

- ☐ **Pre-Warping First:** Never use $\Omega_c = \omega_c/T$ in BLT. Always pre-warp: $\Omega_c = \frac{2}{T} \tan(\omega_c/2)$.
- ☐ **Cross-Term in Denominator:** $(1 - z^{-1})(1 + z^{-1}) = 1 - z^{-2}$. Do not introduce an erroneous z^{-1} cross-term!
- ☐ **Stability Check:** The constant term in the monic denominator ($a_2 = 0.4128$) represents $|z_1 z_2| = r^2$. Since $r = \sqrt{0.4128} \approx 0.64 < 1$, poles are stable.
- ☐ **Zero at Nyquist:** All lowpass filters designed by BLT possess transmission zeros at $z = -1$ ($\omega = \pi$).